Calculus of Variations Midterm Examination

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Name _____ Student ID _____ Class _____

Notation: the functional is denoted by I[x], and a dot denotes differentiation with respect to t. The source image does not specify the duration or point values.

Part I. Variational Problems

Problem 1. Fixed Endpoints

Consider the functional

$$I[x] = \int_0^1 e^{2t}(\dot{x}(t)^2 - x(t)^2)\,dt,$$

$$x(0) = 0, \qquad x(1) = e.$$

(a) Find all functions that can minimize I[x].

(b) Check whether the function obtained is in fact a minimizer. Justify your answer.

Answer Space

Problem 2. Isoperimetric Constraint

Find the function that minimizes the following functional, subject to the stated endpoint and integral constraints. Justify your answer.

$$I[x] = \int_1^2 t^3 \dot{x}(t)^2 \, dt,$$

$$x(1) = 4, \qquad x(2) = 1, \qquad \int_1^2 x(t)\, dt = 2$$

Answer Space

Part II. Theory Questions

Question 1. L Independent of Time

What happens when L is independent of time? Justify your answer.

Question 2. Second Variation

Derive a formula for the second variation.

Question 3. Catenary

Derive and solve the Euler–Lagrange equation for a catenary.

Question 4. Transversality Conditions

What are transversality conditions? Formulate the theorem on the transversality conditions.